

DET-003 | Impossible Travel Detection

ID: DET-003

MITRE ID: T1078

Data Source: SigninLogs

Severity ● Medium

Tactic:Initial Access

Category IDENTITY

ADX MAPPING LOGIC:

Mapping: EventId (numeric) = 'UserID/Session', State = 'Country/Location', StartTime = 'LoginTime'. We use serialize + prev() to detect the same 'user' logging in from two different 'locations' within 60 minutes.

OBJECTIVE: Detect the same user (EventId) appearing from two different locations (State) within 60 minutes — simulating impossible travel. Real Sentinel uses CountryOrRegion changes between successive successful SigninLogs.

KQL QUERY (RUNNABLE ON ADX FREE TIER)

StormEvents

| where StartTime > ago(365d)

| project StartTime, EventId, State, Source

| order by EventId asc, StartTime asc

| serialize

| extend

PrevTime = prev(StartTime, 1),

PrevState = prev(State, 1),

PrevId = prev(EventId, 1)

| where EventId == PrevId // same "user"

| extend TimeDiffMins = datetime_diff('minute', StartTime, PrevTime)

| where TimeDiffMins between (1 .. 60)

and State != PrevState // different "location"

| project

EventId,

CurrentLocation = State,

PreviousLocation = PrevState,

CurrentTime = StartTime,

TimeDiffMins,

Source

| order by TimeDiffMins asc

| take 20

INVESTIGATION STEPS

1. Confirm the user's primary work location.
2. Check if VPN or proxy use is approved for that user.
3. Review what actions were taken after the suspicious login.
4. Contact user via secondary channel (phone) to verify.

RESPONSE ACTIONS

- Suspend the active session token immediately.
- Require step-up MFA for re-authentication.
- Review Azure AD audit logs for changes made in session.
- If unverified, disable the account and escalate to IR.

Output from Above KQL Query:

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | project StartTime, EventId, State, Source
4 | order by EventId asc, StartTime asc
5 | serialize
6 | extend
7   PrevTime = prev(StartTime, 1),
8   PrevState = prev(State, 1),
9   PrevId = prev(EventId, 1)
10 | where EventId == PrevId // same "user"
11 | extend TimeDiffMins = datetime_diff('minute', StartTime, PrevTime)
12 | where TimeDiffMins between (1 .. 60)
13 and State != PrevState // different "location"
14 | project
15   EventId,
16   CurrentLocation = State,
17   PreviousLocation = PrevState,
18   CurrentTime = StartTime,
19   TimeDiffMins,
20   Source
21 | order by TimeDiffMins asc
22 | take 20
```

EventId	CurrentLocation	PreviousLocation	CurrentTime	TimeDiffMins	Source
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No Rows To Show

KQL Detection Query in Sentinel:

SigninLogs

```
| where TimeGenerated > ago(24h)
| where ResultType == "0" // successful logins only
| project TimeGenerated, UserPrincipalName,
IpAddress, Location, CountryOrRegion
| order by UserPrincipalName asc, TimeGenerated asc
| serialize
| extend PrevTime = prev(TimeGenerated, 1),
PrevCountry = prev(CountryOrRegion, 1),
PrevUser = prev(UserPrincipalName, 1)
| where UserPrincipalName == PrevUser
| extend TimeDiffMins = datetime_diff('minute', TimeGenerated, PrevTime)
| where TimeDiffMins < 60 and CountryOrRegion != PrevCountry
| project UserPrincipalName, IAddress,
CurrentCountry = CountryOrRegion, PreviousCountry = PrevCountry,
TimeDiffMins, TimeGenerated
```